

AMMAR RASHEED

Data Engineer | Lakehouse & Cloud ETL Specialist.

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Summary

Data Engineer with strong hands-on experience designing and optimizing batch and real-time data pipelines. Proven expertise in multi-tier (Bronze/Silver/Gold) Lakehouse architectures, distributed data processing with Apache Spark & Databricks, and Oracle database performance tuning. Adept at transforming raw, high-volume data into reliable, production-ready analytics platforms.

Skills

Languages — SQL, PL/SQL, Python (PySpark, Pandas, NumPy), C++

Data Engineering & Cloud — Apache Spark, Apache Kafka, Azure Databricks, Delta Lake, Databricks Auto Loader, Microsoft Fabric

Databases & Warehousing — Oracle (12c/19c), PostgreSQL, MySQL, Enterprise Data Warehousing (Star/Snowflake)

Tools & Tech — Power BI, Toad, DBeaver, SSMS, Git, Azure Cloud

Experience

Data Engineer, (Intern) HiveWorx Pvt. Ltd 12/2025 – Present

- **Oracle PL/SQL & SQL Development:** Authored, refactored, and maintained complex schema-based database objects—including tables, packages, procedures, functions, views, and triggers—in Oracle 12c/19c to migrate 2M+ clinical records with 100% data quality and referential integrity.
- **Performance Tuning & Optimization (PTO):** Analyzed and resolved slow-running queries using native monitoring tools, implementing custom indexing, partitioning, and PL/SQL analytical functions to reduce daily processing times by 40%.
- **Data Modeling & ER Diagrams:** Developed logical and physical data models (ER modeling) and data dictionaries using Toad/DBeaver, establishing star and snowflake schemas that improved downstream reporting query times by 30%.
- **Agile Collaboration & Reviews:** Partnered with software development teams in an Agile Scrum environment; reviewed, modified, and approved proposed database structure changes while participating in code reviews and unit testing to support software releases.
- **Data Quality Assurance:** Implemented CDC-based incremental load strategies (MERGE/UPSERT) in Oracle and built robust validation rules to eliminate data duplication and guarantee data integrity across pipeline executions.
- **Troubleshooting & Real-Time Support:** Provided end-to-end support for critical production systems, independently diagnosing and resolving runtime errors and database issues in real time.
- **Space Management & Infrastructure Maintenance:** Evaluated database systems to perform regular space management and capacity planning, designing metadata-driven orchestrators and audit logging frameworks to track system health.

Projects

Financial Fraud Detection Streaming Platform [🔗](#), *Databricks, Apache Spark Structured Streaming, Apache Kafka (Confluent Cloud), Delta Lake, Python, PostgreSQL, SQL, Databricks Auto Loader, Gmail SMTP, Git*

- **Streaming Architecture:** Architected an end-to-end real-time streaming pipeline using Kafka and Spark Structured Streaming, processing 1,000+ transactions per minute (TPM) with sub-second latency.
- **Data Ingestion & Enrichment:** Streamlined ingestion using Databricks Auto Loader, dynamically enriching streaming transactions with customer master data from PostgreSQL with 99.8% matching accuracy.
- **Alerting & Logic:** Coded custom rule-based fraud detection logic that automatically identifies high-value anomalies and triggers automated email alerts via SMTP within 2 seconds of detection.
- **Real-time Visualization:** Built interactive dashboards to monitor transaction volume, fraud alerts, and key operational KPIs, lowering incident response times by 35%.

Enterprise-Sales-Data-Warehouse-and-Analytics-Platform [🔗](#)

- **ETL Optimization:** Programmed robust stored procedures with custom transaction management to orchestrate ETL workflows across 25+ dimension and fact tables, improving batch processing speed by 30%.
- **Performance Monitoring:** Created a database monitoring solution to track query runtimes and row counts, reducing diagnostic time for slow-running procedures by 50%.
- **Exception Handling:** Implemented automated error-handling mechanisms using TRY...CATCH blocks, reducing manual pipeline debugging effort by 40%.
- **BI Reporting:** Delivered 5+ Power BI dashboards visualizing 15+ retail KPIs, providing stakeholders with instant visibility into business trends and operational metrics.

Healthcare Enterprise Lakehouse — Medallion Architecture, AI & Analytics Platform,

IHHN- Indus Hospital & Health Network

- **Lakehouse Architecture:** Architected a metadata-enabled data lakehouse for a 7+ campus hospital system, establishing a structured data flow from raw ingestion to curated analytical datasets.
- **Scalable ETL & Compliance:** Built scalable, HIPAA-compliant ETL pipelines to process 10M+ patient and clinical records while strictly enforcing data privacy and security policies.
- **Data Mart Design:** Modeled and deployed a clinical data store containing 50+ dimension and fact tables, simplifying multi-facility reporting and analytics.
- **Clinical Dashboards:** Developed performance dashboards displaying 100+ clinical and operational KPIs (e.g., patient flow, emergency metrics), enabling administrators to optimize resource allocation.

Education

Bachelor of Science in Computer Science, SZABIST ISB

02/2021 – 02/2025 | ISLAMABAD, PAKISTAN

Courses

Google Data Analytics, Coursera

Python for Data Analytics, Udemy

Azure Data Engineer Associate, Udemy